

WEATHER ELEMENTS / INDIA JET STREAM-WESTERN DISTURBANCES

WEATHER CAUSATION CHAIN → HEAT-BUDGET LEDGER → STABILITY PROFILES → WIND-FORCE TRIANGLE → PLANETARY CIRCULATION RAIL → HUMIDITY-TO-CLOUD CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 WEATHER CAUSATION CHAIN

WEATHER CAUSATION CHAIN

SURFACE AND ATMOSPHERIC HEATING

TEMPERATURE GRADIENT

PRESSURE GRADIENT

MOISTURE TRANSPORT AND CONVERGENCE

LIFT + SATURATION

CLOUD AND PRECIPITATION

MUST REMEMBER

INSOLATION → surface and atmospheric heating

TEMPERATURE GRADIENT → pressure gradient

WIND → moisture transport and convergence

LIFT + SATURATION → cloud and precipitation

MUST REMEMBER: Weather elements connect radiation balance, temperature, pressure-gradient force, Coriolis/friction, humidity, stability, condensation, clouds

CORE CLAIM

- INSOLATION → surface and atmospheric heating
- TEMPERATURE GRADIENT → pressure gradient

MECHANISM

- WIND → moisture transport and convergence
- LIFT + SATURATION → cloud and precipitation

CONSEQUENCE / LIMIT

- MUST REMEMBER: Weather elements connect radiation balance, temperature, pressure-gradient force, Coriolis/friction, humidity, stability, condensation, clouds and precipitation
- upper-air Rossby waves, subtropical jet and western-disturbance tracks anchor Indian seasonality.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: upper-air Rossby waves, subtropical jet and western-disturbance tracks anchor Indian seasonality.

01

STAGE 01 HEAT-BUDGET LEDGER

HEAT-BUDGET LEDGER

INCOMING SHORTWAVE

REFLECTED + ABSORBED ENERGY

LONGWAVE + SENSIBLE + LATENT HEAT

TEMPERATURE AND CIRCULATION RESPONSE

SUN → incoming shortwave

SURFACE → reflected + absorbed energy

EARTH → longwave + sensible + latent heat

BALANCE → temperature and circulation response

CORE CLAIM

- SUN → incoming shortwave
- SURFACE → reflected + absorbed energy

MECHANISM

- EARTH → longwave + sensible + latent heat
- BALANCE → temperature and circulation response

CONSEQUENCE / LIMIT

- SUN → incoming shortwave
- SURFACE → reflected + absorbed energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARTH → longwave + sensible + latent heat.

02

STAGE 02 STABILITY PROFILES

STABILITY PROFILES

NORMAL TENDENCY

TEMPERATURE FALLS UPWARD

TEMPERATURE RISES THROUGH LAYER

STABLE LAYER

VERTICAL MIXING SUPPRESSED

SURFACE INVERSION

FOG AND POLLUTION TRAPPING

CORE CLAIM

NORMAL TENDENCY → temperature falls upward
INVERSION → temperature rises through layer

MECHANISM

STABLE LAYER → vertical mixing suppressed
SURFACE INVERSION → fog and pollution trapping

CONSEQUENCE / LIMIT

NORMAL TENDENCY → temperature falls upward
INVERSION → temperature rises through layer

CORE CLAIM

- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

MECHANISM

- STABLE LAYER → vertical mixing suppressed
- SURFACE INVERSION → fog and pollution trapping

CONSEQUENCE / LIMIT

- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE LAYER → vertical mixing suppressed; SURFACE INVERSION → fog and pollution trapping.

03

STAGE 03 WIND-FORCE TRIANGLE

WIND-FORCE TRIANGLE

PRESSURE GRADIENT

STARTS ACCELERATION

RIGHT NORTH

LEFT SOUTH

SLOWS AND CROSSES ISOBARS

NEAR-GEOSTROPHIC PARALLEL FLOW

PRESSURE GRADIENT → starts acceleration

CORIOLIS → right north / left south

FRICTION → slows and crosses isobars

ALOFT → near-geostrophic parallel flow

CORE CLAIM

- PRESSURE GRADIENT → starts acceleration

MECHANISM

- FRICTION → slows and crosses isobars

CONSEQUENCE / LIMIT

- PRESSURE GRADIENT → starts acceleration

03

CORE CLAIM

- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

MECHANISM

- STABLE LAYER → vertical mixing suppressed
- SURFACE INVERSION → fog and pollution trapping

CONSEQUENCE / LIMIT

- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE LAYER → vertical mixing suppressed; SURFACE INVERSION → fog and pollution trapping.

04

STAGE 03

WIND-FORCE TRIANGLE

WIND-FORCE TRIANGLE

PRESSURE GRADIENT

STARTS ACCELERATION

RIGHT NORTH

LEFT SOUTH

SLOWS AND CROSSES ISOBARS

NEAR-GEOSTROPHIC PARALLEL FLOW

PRESSURE GRADIENT → starts acceleration

CORIOLIS → right north / left south

FRICTION → slows and crosses isobars

ALOFT → near-geostrophic parallel flow

CORE CLAIM

- PRESSURE GRADIENT → starts acceleration
- CORIOLIS → right north / left south

MECHANISM

- FRICTION → slows and crosses isobars
- ALOFT → near-geostrophic parallel flow

CONSEQUENCE / LIMIT

- PRESSURE GRADIENT → starts acceleration
- CORIOLIS → right north / left south

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRICTION → slows and crosses isobars; ALOFT → near-geostrophic parallel flow.

05

STAGE 04

PLANETARY CIRCULATION RAIL

PLANETARY CIRCULATION RAIL

EQUATORIAL LOW

CONVERGENCE AND ASCENT

SUBTROPICAL HIGH

DESCENT AND TRADES

SUBPOLAR LOW

WESTERLY-POLAR INTERACTION

POLAR HIGH

EQUATORIAL LOW → convergence and ascent

SUBTROPICAL HIGH → descent and trades

SUBPOLAR LOW → westerly-polar interaction

POLAR HIGH → cold outflow; belts migrate

CORE CLAIM

- EQUATORIAL LOW → convergence and ascent
- SUBTROPICAL HIGH → descent and trades

MECHANISM

- SUBPOLAR LOW → westerly-polar interaction
- POLAR HIGH → cold outflow; belts migrate

CONSEQUENCE / LIMIT

- EQUATORIAL LOW → convergence and ascent
- SUBTROPICAL HIGH → descent and trades

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUBPOLAR LOW → westerly-polar interaction; POLAR HIGH → cold outflow; belts migrate.

06

STAGE 05

HUMIDITY-TO-CLOUD CHAIN

HUMIDITY-TO-CLOUD CHAIN

MOISTURE CONTENT + TEMPERATURE

RELATIVE HUMIDITY

COOLING TO SATURATION

DEW POINT REACHED

NUCLEI + LIFT

DROPLETS OR ICE CRYSTALS

GROWTH + FALL SPEED

MOISTURE CONTENT + TEMPERATURE → relative humidity

COOLING TO SATURATION → dew point reached

NUCLEI + LIFT → droplets or ice crystals

GROWTH + FALL SPEED → precipitation

CORE CLAIM

- MOISTURE CONTENT + TEMPERATURE → relative humidity
- COOLING TO SATURATION → dew point reached

MECHANISM

- NUCLEI + LIFT → droplets or ice crystals
- GROWTH + FALL SPEED → precipitation

CONSEQUENCE / LIMIT

- MOISTURE CONTENT + TEMPERATURE → relative humidity
- COOLING TO SATURATION → dew point reached

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUCLEI + LIFT → droplets or ice crystals.

07

STAGE 06

RAINFALL COMPARISON

RAINFALL COMPARISON

BUOYANT THERMAL ASCENT

FORCED RELIEF ASCENT

CONVERGENT DYNAMIC ASCENT

REAL EVENT

MECHANISMS MAY COMBINE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CONVECTIONAL → buoyant thermal ascent	FRONTAL / CYCLONIC → convergent dynamic ascent	CONVECTIONAL → buoyant thermal ascent
OROGRAPHIC → forced relief ascent	REAL EVENT → mechanisms may combine	OROGRAPHIC → forced relief ascent

CORE CLAIM

- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

MECHANISM

- FRONTAL / CYCLONIC → convergent dynamic ascent
- REAL EVENT → mechanisms may combine

CONSEQUENCE / LIMIT

- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRONTAL / CYCLONIC → convergent dynamic ascent; REAL EVENT → mechanisms may combine.

WEATHER ELEMENTS / INDIA JET STREAM-WESTERN DISTURBANCES • TILE 2/4 • same master rows 2430-5664 of 10524 • continues below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUCLEI + LIFT → droplets or ice crystals.

06

STAGE 06

RAINFALL COMPARISON

RAINFALL COMPARISON

BUOYANT THERMAL ASCENT

FORCED RELIEF ASCENT

CONVERGENT DYNAMIC ASCENT

REAL EVENT

MECHANISMS MAY COMBINE

CORE CLAIM

CONVECTIONAL → buoyant thermal ascent

OROGRAPHIC → forced relief ascent

MECHANISM

FRONTAL / CYCLONIC → convergent dynamic ascent

REAL EVENT → mechanisms may combine

CONSEQUENCE / LIMIT

CONVECTIONAL → buoyant thermal ascent

OROGRAPHIC → forced relief ascent

CORE CLAIM

• CONVECTIONAL → buoyant thermal ascent

• OROGRAPHIC → forced relief ascent

MECHANISM

• FRONTAL / CYCLONIC → convergent dynamic ascent

• REAL EVENT → mechanisms may combine

CONSEQUENCE / LIMIT

• CONVECTIONAL → buoyant thermal ascent

• OROGRAPHIC → forced relief ascent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRONTAL / CYCLONIC → convergent dynamic ascent; REAL EVENT → mechanisms may combine.

07

STAGE 07

TROPOSPHERE SECTION

TROPOSPHERE SECTION

MOISTURE, FRICTION AND HEATING

MID LEVELS

CLOUDS, TROUGHS AND STABILITY

UPPER TROPOSPHERE

JETS AND DIVERGENCE

TROPICS DEEPER

POLES SHALLOWER

SURFACE → moisture, friction and heating

MID LEVELS → clouds, troughs and stability

UPPER TROPOSPHERE → jets and divergence

TROPICS DEEPER / POLES SHALLOWER

CORE CLAIM

• SURFACE → moisture, friction and heating

• MID LEVELS → clouds, troughs and stability

MECHANISM

• UPPER TROPOSPHERE → jets and divergence

• TROPICS DEEPER / POLES SHALLOWER

CONSEQUENCE / LIMIT

• SURFACE → moisture, friction and heating

• MID LEVELS → clouds, troughs and stability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UPPER TROPOSPHERE → jets and divergence; TROPICS DEEPER / POLES SHALLOWER.

08

STAGE 08

JET-STREAM ENGINE

JET-STREAM ENGINE

HORIZONTAL TEMPERATURE GRADIENT

PRESSURE GRADIENT ALOFT

CORIOLIS BALANCE

STRONG NEAR-ZONAL WIND

ROSSBY WAVES

RIDGES, TROUGHS AND STEERING

HORIZONTAL TEMPERATURE GRADIENT

HEIGHT / PRESSURE GRADIENT ALOFT

CORIOLIS BALANCE → strong near-zonal wind

ROSSBY WAVES → ridges, troughs and steering

CORE CLAIM

• HORIZONTAL TEMPERATURE GRADIENT

• HEIGHT / PRESSURE GRADIENT ALOFT

MECHANISM

• CORIOLIS BALANCE → strong near-zonal wind

• ROSSBY WAVES → ridges, troughs and steering

CONSEQUENCE / LIMIT

• HORIZONTAL TEMPERATURE GRADIENT

• HEIGHT / PRESSURE GRADIENT ALOFT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CORIOLIS BALANCE → strong near-zonal wind; ROSSBY WAVES → ridges, troughs and steering.

09

STAGE 09

INDIA SEASONAL JETS

INDIA SEASONAL JETS

STWJ SOUTH OF HIMALAYA SECTOR

STEERS WESTERN DISTURBANCES EAST

TROPICAL EASTERLY JET DEVELOPS

JETS ARE PARTS OF WIDER CIRCULATION

CLOSE DISTINCTION

HUMIDITY, RELATIVE HUMIDITY AND DEW POINT DIFFER; WEATHER DIFFERS

WINTER → STWJ south of Himalaya sector

STWJ → steers western disturbances east

SUMMER → tropical easterly jet develops

RULE → jets are parts of wider circulation

CORE CLAIM

• WINTER → STWJ south of Himalaya sector

• STWJ → steers western disturbances east

MECHANISM

• SUMMER → tropical easterly jet develops

• RULE → jets are parts of wider circulation

CONSEQUENCE / LIMIT

• CLOSE DISTINCTION: Humidity, relative humidity and dew point differ

• weather differs from climate, jet stream from surface wind, and a western disturbance is an eastward-moving extratropical system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → jets are parts of wider circulation.

09

STAGE 09

INDIA SEASONAL JETS

INDIA SEASONAL JETS

STWJ SOUTH OF HIMALAYA SECTOR

STEERS WESTERN DISTURBANCES EAST

TROPICAL EASTERLY JET DEVELOPS

JETS ARE PARTS OF WIDER CIRCULATION

CLOSE DISTINCTION

HUMIDITY, RELATIVE HUMIDITY AND DEW POINT DIFFER; WEATHER DIFFERS

WINTER → STWJ south of Himalaya sector

STWJ → steers western disturbances east

SUMMER → tropical easterly jet develops

RULE → jets are parts of wider circulation

CORE CLAIM

- WINTER → STWJ south of Himalaya sector
- STWJ → steers western disturbances east

MECHANISM

- SUMMER → tropical easterly jet develops
- RULE → jets are parts of wider circulation

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Humidity, relative humidity and dew point differ
- weather differs from climate, jet stream from surface wind, and a western disturbance is an eastward-moving extratropical system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → jets are parts of wider circulation.

10

STAGE 10

WESTERN-DISTURBANCE CHAIN

WESTERN-DISTURBANCE CHAIN

UPPER TROUGH

EASTWARD WESTERLY MOTION

MOISTURE + DYNAMICAL LIFT

CLOUD AND RAIN/SNOW

HIMALAYA + PLAINS

TERRAIN-SPECIFIC IMPACTS

BENEFICIAL OR DAMAGING BY TIMING/INTENSITY

UPPER TROUGH / LOW → eastward westerly motion

MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

HIMALAYA + PLAINS → terrain-specific impacts

OUTCOME → beneficial or damaging by timing/intensity

EVIDENCE LIMIT: Jet position is one control among topography, blocking, moisture supply and synoptic evolution

CORE CLAIM

- UPPER TROUGH / LOW → eastward westerly motion
- MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

MECHANISM

- HIMALAYA + PLAINS → terrain-specific impacts
- OUTCOME → beneficial or damaging by timing/intensity

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: Jet position is one control among topography, blocking, moisture supply and synoptic evolution
- event attribution and seasonal forecasts require dated agency evidence and probabilistic language.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Jet position is one control among topography, blocking, moisture supply and synoptic evolution; event attribution and seasonal forecasts require dated agency evidence and probabilistic language.

11

STAGE 11

FORECAST EVIDENCE LADDER

FORECAST EVIDENCE LADDER

PILOT BALLOON

UPPER-AIR STATE

SATELLITE + RADAR

CLOUD AND PRECIPITATION

NWP ENSEMBLE

SCENARIOS AND UNCERTAINTY

ACTION; LATER OBSERVATIONS VERIFY OUTCOME

RADIOSONDE / PILOT BALLOON → upper-air state

SATELLITE + RADAR → cloud and precipitation

NWP ENSEMBLE → scenarios and uncertainty

WARNING → action; later observations verify outcome

CORE CLAIM

- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

MECHANISM

- NWP ENSEMBLE → scenarios and uncertainty
- WARNING → action; later observations verify outcome

CONSEQUENCE / LIMIT

- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NWP ENSEMBLE → scenarios and uncertainty; WARNING → action; later observations verify outcome.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

JET STREAMS ARE FAST, NARROW UPPER-TROPOSPHERIC WIND BANDS.

STWJ LIES SOUTH OF THE HIMALAYAS IN WINTER AND

NORTHWARD WITHDRAWAL OF STWJ IS A PRE-CONDITION FOR SW

TEJ DEVELOPS AS PART OF THE SUMMER MONSOON'S UPPER-AIR

WESTERN DISTURBANCES ARE EXTRA-TROPICAL, NON-MONSOONAL LOW-PRESSURE SYSTEMS FROM THE

WDS BRING WINTER/EARLY-SPRING RAIN TO PUNJAB, HARYANA, DELHI AND

THEY ARE VITAL FOR RABI CROPS SUCH AS WHEAT

OPTIONAL SOURCE DEPTH

- Jet streams are fast, narrow upper-tropospheric wind bands.
- STWJ lies south of the Himalayas in winter and acts as the conveyor for Western Disturbances.
- Northward withdrawal of STWJ is a pre-condition for SW monsoon onset.
- TEJ develops as part of the summer monsoon's upper-air circulation. Tibetan/Asian heating matters,

ADVANCED DEBATE

- Western Disturbances are extra-tropical, non-monsoonal low-pressure systems from the Mediterranean moving east via Iran-Afghanistan-Pakistan.
- WDs bring winter/early-spring rain to Punjab, Haryana, Delhi and UP and snow to J&K, HP and Uttarakhand
- they are vital for Rabi crops such as wheat, mustard and gram.
- In May–June, intense heating over north-west India supports the hot, dry Loo. Its timing and strength

LEGACY / NUANCE

- Altitude lowers temperature by about 6.5 deg C/km in the notes
- the Himalayan barrier gives India warmer winters than comparable latitudes by blocking Central Asian cold air.
- STWJ steers Western Disturbances in winter.
- STWJ's northward shift is essential for SW monsoon onset.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.